

Module 4

Problem Statement:

You have to create an inheritance among **Father**→**Son** →**GrandSon** class. The **Father** class has the following data members

```
class Father
{
    private:
        int money;
    protected:
        int gold;
    public:
        int land;
};
```

Now write the **Son** and **GrandSon** classes with **private/protected/public** access modifier and do the following:

- i) Try to access **money**, **gold** and **land** from Son class
- ii) Try to access **money**, **gold** and **land** from GrandSon class
- iii) Find the values of money, gold and land when different access modifier is used in the following table

Code-1:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define endl "\n"
#define newl cout<<endl
class Father
{
    private:
        int money=1000;
    protected:
        int gold=10000;
    public:
        int land=1e8;
        Father()
        {

        }
};
class child: public Father
{
    public:
        child()
        {

        }
    void show()
    {
        cout<<money;
        newl;
        cout<<gold;
        newl;
        cout<<land;
        newl;
    }
};
class GrandSon: public child
{
    public:
        GrandSon()
```

```

{
}
void show()
{
    cout<<money;
    newl;
    cout<<gold;
    newl;
    cout<<land;
    newl;
}
};
int main()
{
    GrandSon g;
    g.show();
    child c;
    c.show();
}

```

Output-1:

C:\Users\MD...		In member function 'void child::show()':
C:\Users\MD...	29	error: 'int Father::money' is private within this context
C:\Users\MD...	10	note: declared private here
C:\Users\MD...		In member function 'void GrandSon::show()':
C:\Users\MD...	46	error: 'int Father::money' is private within this context
C:\Users\MD...	10	note: declared private here

Code-2:

```

// Header File from Code-1
class Father
{
    // from code-1 statement
};
class Son: protected Father
{
    // from code-1 statement
};
class GrandSon: public Son
{
    // from code-1 statement
};
int main()
{
    // from code-1 statement
}

```

Output-2:

C:\Users\MD...		In member function 'void Son::show()':
C:\Users\MD...	29	error: 'int Father::money' is private within this context
C:\Users\MD...	10	note: declared private here
C:\Users\MD...		In member function 'void GrandSon::show()':
C:\Users\MD...	46	error: 'int Father::money' is private within this context
C:\Users\MD...	10	note: declared private here

Code-3:

```

// Header File from Code-1
class Father
{
    // from code-1 statement
};
class Son: private Father
{

```

```

// from code-1 statement
};
class GrandSon: public Son
{
    // from code-1 statement
};
int main()
{
    // from code-1 statement
}

```

Output-3:

C:\Users\MD...		In member function 'void Son::show()':
C:\Users\MD...	29	error: 'int Father::money' is private within this context
C:\Users\MD...	10	note: declared private here
C:\Users\MD...		In member function 'void GrandSon::show()':
C:\Users\MD...	46	error: 'int Father::money' is private within this context
C:\Users\MD...	10	note: declared private here
C:\Users\MD...	10	note: declared private here
C:\Users\MD...	48	error: 'int Father::gold' is protected within this context
C:\Users\MD...	12	note: declared protected here
C:\Users\MD...	12	note: declared protected here
C:\Users\MD...	12	note: declared protected here
C:\Users\MD...	50	error: 'int Father::land' is inaccessible within this context
C:\Users\MD...	14	note: declared here

Code-4:

```

// Header File from Code-1
class Father
{
    // from code-1 statement
};
class Son: public Father
{
    // from code-1 statement
};
class GrandSon: protected Son
{
    // from code-1 statement
};
int main()
{
    // from code-1 statement
}

```

Output-4:

C:\Users\MD...		In member function 'void Son::show()':
C:\Users\MD...	29	error: 'int Father::money' is private within this context
C:\Users\MD...	10	note: declared private here
C:\Users\MD...		In member function 'void GrandSon::show()':
C:\Users\MD...	46	error: 'int Father::money' is private within this context
C:\Users\MD...	10	note: declared private here

Code-5:

```

// Header File from Code-1
class Father
{
    // from code-1 statement
};
class Son: protected Father
{
    // from code-1 statement
};
class GrandSon: protected Son
{
    // from code-1 statement
};
int main()
{
    // from code-1 statement
}

```

Output-5:

C:\Users\MD...		In member function 'void Son::show()':
C:\Users\MD...	29	error: 'int Father::money' is private within this context
C:\Users\MD...	10	note: declared private here
C:\Users\MD...		In member function 'void GrandSon::show()':
C:\Users\MD...	46	error: 'int Father::money' is private within this context
C:\Users\MD...	10	note: declared private here

Code-6:

```
// Header File from Code-1
class Father
{
    // from code-1 statement
};
class Son: private Father
{
    // from code-1 statement
};
class GrandSon: protected Son
{
    // from code-1 statement
};
int main()
{
    // from code-1 statement
}
```

Output-6:

C:\Users\MD...		In member function 'void Son::show()':
C:\Users\MD...	29	error: 'int Father::money' is private within this context
C:\Users\MD...	10	note: declared private here
C:\Users\MD...		In member function 'void GrandSon::show()':
C:\Users\MD...	46	error: 'int Father::money' is private within this context
C:\Users\MD...	10	note: declared private here
C:\Users\MD...	10	note: declared private here
C:\Users\MD...	48	error: 'int Father::gold' is protected within this context
C:\Users\MD...	12	note: declared protected here
C:\Users\MD...	50	error: 'int Father::land' is inaccessible within this context

Code7:

```
// Header File from Code-1
class Father
{
    // from code-1 statement
};
class Son: public Father
{
    // from code-1 statement
};
class GrandSon: private Son
{
    // from code-1 statement
};
int main()
{
    // from code-1 statement
}
```

Output-7:

C:\Users\MD...		In member function 'void Son::show()':
C:\Users\MD...	29	error: 'int Father::money' is private within this context
C:\Users\MD...	10	note: declared private here
C:\Users\MD...		In member function 'void GrandSon::show()':
C:\Users\MD...	46	error: 'int Father::money' is private within this context
C:\Users\MD...	10	note: declared private here

Code-8:

```
// Header File from Code-1
class Father
{
    // from code-1 statement
};
```

```

class Son: protected Father
{
    // from code-1 statement
};
class GrandSon: private Son
{
    // from code-1 statement
};
int main()
{
    // from code-1 statement
}

```

Output-8:

C:\Users\MD...		In member function 'void Son::show()':
C:\Users\MD...	29	error: 'int Father::money' is private within this context
C:\Users\MD...	10	note: declared private here
C:\Users\MD...		In member function 'void GrandSon::show()':
C:\Users\MD...	46	error: 'int Father::money' is private within this context
C:\Users\MD...	10	note: declared private here

Code-9:

```

// Header File from Code-1
class Father
{
    // from code-1 statement
};
class Son: private Father
{
    // from code-1 statement
};
class GrandSon: private Son
{
    // from code-1 statement
};
int main()
{
    // from code-1 statement
}

```

Output-9:

C:\Users\MD...		In member function 'void Son::show()':
C:\Users\MD...	29	error: 'int Father::money' is private within this context
C:\Users\MD...	10	note: declared private here
C:\Users\MD...		In member function 'void GrandSon::show()':
C:\Users\MD...	46	error: 'int Father::money' is private within this context
C:\Users\MD...	10	note: declared private here
C:\Users\MD...	10	note: declared private here
C:\Users\MD...	48	error: 'int Father::gold' is protected within this context
C:\Users\MD...	12	note: declared protected here
C:\Users\MD...	50	error: 'int Father::land' is inaccessible within this context

So, the **Inheritance** table is below:

Class		In Son class			In GrandSon class		
Son	GrandSon	money	gold	land	money	gold	land
public	public	No	Yes	Yes	No	Yes	Yes
protected	public	No	Yes	Yes	No	Yes	Yes
private	public	No	Yes	Yes	No	No	No
public	protected	No	Yes	Yes	No	Yes	Yes
protected	protected	No	Yes	Yes	No	Yes	Yes
private	protected	No	Yes	Yes	No	No	No
public	private	No	Yes	Yes	No	Yes	Yes
protected	private	No	Yes	Yes	No	Yes	Yes
private	private	No	Yes	Yes	No	No	No